

# ISyE 6669

## Final Exam Practice

Instructor: Prof. Shabbir Ahmed and Prof. Andy Sun

**Problem 1.** Which of the following sets is convex for any positive integer  $n$ ?

A  $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i \geq -1\}.$

B  $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i = -1\}.$

C  $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i \leq -1\}.$

**Problem 2.** Which of the following functions is convex on the specified domain?

A  $f(x) = 5 - |x|$  on  $x \leq 0$

B  $f(x) = x - |x|$  on  $x \in [-7, 1]$

C  $f(x) = -|x|$  on  $x \in \mathbb{R}$

**Problem 3.** Which of the following statements is true about the extreme points and extreme rays?

A Any polyhedron has an extreme ray.

B If a nonempty polyhedron does not have any extreme point, then it is unbounded.

C For a nonempty polyhedron, any point in the polyhedron can be written as the non-zero sum of at least two different extreme points and some extreme rays.

**Problem 4.** Consider a constrained optimization problem

$$\begin{aligned} \min & f(x) \\ \text{s.t. } & g(x) \geq 0, \\ & h(x) = 0, \\ & x \in X. \end{aligned}$$

where  $h(x) \geq 0$  for all  $x \in X$ . Which of the following problems is a relaxation of this problem?

A  $\min\{f(x) + h(x) : g(x) \geq -1, x \in X\}$

B  $\max\{f(x) + g(x) : h(x) = 10, x \in X\}$

C  $\min\{-g(x) + h(x) : x \in X\}$

**Problem 5.** Consider the following linear program for  $(x_1, x_2, x_3) \in \mathbb{R}^3$ :

$$\begin{aligned} \min \quad & 3x_1 + 5x_2 + 7x_3 \\ \text{s.t.} \quad & -x_1 + 3x_2 = 5, \\ & 2x_1 - x_2 + 3x_3 \geq 6, \\ & x_3 \leq 7, \\ & x_1 \geq 0, x_2 \leq 0, x_3 \text{ free.} \end{aligned}$$

The dual of the above linear program can be written as:

A

$$\begin{aligned} \max \quad & 5y_1 + 6y_2 + 7y_3 \\ \text{s.t.} \quad & -y_1 + 2y_2 \leq 3, \\ & 3y_1 - y_2 \geq 5, \\ & 3y_2 + y_3 = 7, \\ & y_1 \text{ free}, y_2 \leq 0, y_3 \geq 0. \end{aligned}$$

B

$$\begin{aligned} \max \quad & 5y_1 + 6y_2 + 7y_3 \\ \text{s.t.} \quad & -y_1 + 2y_2 \geq 3, \\ & 3y_1 - y_2 \leq 5, \\ & 3y_2 + y_3 = 7, \\ & y_1 \text{ free}, y_2 \geq 0, y_3 \leq 0. \end{aligned}$$

C

$$\begin{aligned} \max \quad & 5y_1 + 6y_2 + 7y_3 \\ \text{s.t.} \quad & -y_1 + 2y_2 \leq 3, \\ & 3y_1 - y_2 \geq 5, \\ & 3y_2 + y_3 = 7, \\ & y_1 \text{ free}, y_2 \geq 0, y_3 \leq 0. \end{aligned}$$

**Problem 6.** Consider the following linear program:

$$(P) \quad \min \quad c^T x \\ \text{s.t.} \quad Ax \leq 0, \\ 0 \leq x \leq d$$

Select all correct statements.

- A The problem (P) is possible to be infeasible.
- B If the problem (P) is unbounded, then its dual is unbounded.
- C If the problem (P) has a finite optimal, then its dual has a finite optimal.

**Problem 7.** Consider the robust linear constraint for  $x \in \mathbb{R}^n$ :

$$a_1^\top x + a_2^\top y \geq b, \quad \forall a \in \mathcal{A} = \{a_1, a_2 \in [0, 1]^n : C_1 a_1 \leq d_1, C_2 a_2 \leq d_2\}$$

Which of the following is the correct reformulation.

- A  $C_1 a_1 + C_2 a_2 - d_1 - d_2 \geq b$ .
- B  $d_1^\top z + d_2^\top w + e^\top u + e^\top v + b \leq 0, \quad x + C_1^\top z + u \geq 0, \quad y + C_2^\top w + v \geq 0, \quad z, w, u, v \geq 0$ , where  $e = (1, \dots, 1)^\top \in \mathbb{R}^n$ .
- C  $\sum_{i=1}^n x_i + y_i \geq b$ .

**Problem 8.** Consider a simple cutting stock problem, where a big roll has width 7 and demand takes widths 3 and 2. How many different columns are there in the cutting stock problem formulation?

- A 6
- B 7
- C 10

**Problem 9.** Which of the following statements are true about solving a linear program by the column generation algorithm?

- A The column generation algorithm will continue if all the columns are added
- B To start solving the problem through the column generation algorithm, we need to know all the possible patterns

C The optimal solution of Restricted Master Problem in the column generation algorithm is always a basic feasible solution of Master Problem

**Problem 10.** For the polytope  $P = [-1, 0]^n$ , the extreme point representation of any point  $x \in P$  can be written as  $x = \sum_{i=1}^n -\lambda_i e_i$ ,  $\sum_{i=1}^n \lambda_i = 1$ ,  $\lambda_i \geq 0$ ,  $i = 1, \dots, n$  where  $e_i$  is a vector with the  $i$ -th component being 1 and all other components being 0.

A True

B False

**Problem 11.** Consider the following linear program:

$$\min \quad 3x_1 + 5x_2 + 2x_3 + 2x_4 \quad (1)$$

$$\text{s.t.} \quad x_1 - 3x_2 + 2x_3 + x_4 \leq 1 \quad (2)$$

$$x_1 + x_2 \leq 1 \quad (3)$$

$$-x_1 + x_2 \leq 1 \quad (4)$$

$$x_2 \geq 0 \quad (5)$$

$$0 \leq x_3 \leq 2 \quad (6)$$

$$0 \leq x_4 \leq 1 \quad (7)$$

Denote the polyhedron for variables  $x_1$  and  $x_2$  defined by constraints (3)-(5) as  $P_1$ , and denote the polyhedron for  $x_3$  and  $x_4$  defined by constraints (6)-(7) as  $P_2$ . Use two extreme points of  $P_1 : (x_1^1, x_2^1) = (0, 1)$  and  $(x_1^2, x_2^2) = (1, 0)$ , and two extreme points of  $P_2 : (x_3^1, x_4^1) = (0, 0)$  and  $(x_3^2, x_4^2) = (2, 0)$ . Which of the following is a correct restricted master problem of the linear program?

A

$$\min \quad 5\lambda_1 + 2\lambda_2 + 4\beta_2$$

$$\text{s.t.} \quad -3\lambda_1 + \lambda_2 + 4\beta_2 \leq 1$$

$$\lambda_1 + \lambda_2 \leq 1$$

$$\beta_1 + \beta_2 = 1$$

$$\lambda_1, \lambda_2 \geq 0$$

$$\beta_1, \beta_2 \geq 0.$$

B

$$\min \quad 5\lambda_1 + 2\lambda_2 + 2\beta_2$$

$$\text{s.t.} \quad -2\lambda_1 + \lambda_2 + 4\beta_2 \leq 1$$

$$\lambda_1 + \lambda_2 = 1$$

$$\beta_1 + \beta_2 = 1$$

$$\lambda_1, \lambda_2 \geq 0$$

$$\beta_1, \beta_2 \geq 0.$$

C

$$\begin{array}{ll}\min & 5\lambda_1 + 3\lambda_2 + 4\beta_2 \\ \text{s.t.} & -3\lambda_1 + \lambda_2 + 4\beta_2 \leq 1 \\ & \lambda_1 + \lambda_2 = 1 \\ & \beta_1 + \beta_2 = 1 \\ & \lambda_1, \lambda_2 \geq 0 \\ & \beta_1, \beta_2 \geq 0.\end{array}$$

**Problem 12.** Which of the following points is the closest to  $(4, 3, 2, 1)^\top \in \mathbb{R}^4$  in Euclidean norm?

- A  $(3, 1, 2, 4)^\top$
- B  $(4, 2, 1, 3)^\top$
- C  $(3, 4, 1, 2)^\top$

**Problem 13.** If we want to solve a least square problem  $\min_x \|Ax - b\|_2^2$  where  $A = \begin{pmatrix} 4 & 3 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$  and  $b = (4, 6, 2)^\top$ , then

- A The optimal solution is  $x = (2, -4/3)$ .
- B The optimal solution is  $x = (2, 4/3)$ .
- C The optimal solution is  $x = (-2, 4/3)$ .
- D The optimal solution is  $x = (-2, -4/3)$ .

**Problem 14.** If  $A \in \mathbb{R}^{3 \times 3}$  is a symmetric matrix with singular values  $\sigma_1 = 4$ ,  $\sigma_2 = 1$ , and  $\sigma_3 = 0$ , which of the following statements is true about the eigenvalues of  $A$ ?

- A The matrix  $A$  may not have eigenvalues.
- B The eigenvalues of  $A$  can be  $\lambda_1 = 2$ ,  $\lambda_2 = 1$ , and  $\lambda_3 = 0$ .
- C The eigenvalues of  $A$  can be  $\lambda_1 = 4$ ,  $\lambda_2 = 1$ , and  $\lambda_3 = 0$ .

**Problem 15.** Consider the least square regression problem

$$z_1 = \min_{\beta_1, \dots, \beta_n} \sum_{i=1}^N (y_i - \sum_{j=1}^n \beta_j x_{ij})^2$$

and the absolute deviation regression problem defined as

$$z_2 = \min_{\beta_1, \dots, \beta_n} \sum_{i=1}^N |y_i - \sum_{j=1}^n \beta_j x_{ij}|$$

where  $x_{ij}(i = 1, \dots, N; j = 1, \dots, n); y_i(i = 1, \dots, N)$  are training data. Which of the following statements is true?

- A  $z_1$  and  $z_2$  are always equal.
- B We cannot solve the absolute deviation regression problem by linear programming.
- C The optimal solutions can be different.

**Problem 16.** Is  $\begin{pmatrix} 3 \\ -3 \\ 5 \end{pmatrix} \preceq_{\mathbb{R}_+^3} \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ ?

- A Yes
- B No

**Problem 17.** Which of the following cannot be formulated as a second-order cone constraint?

- A  $x^2 + y^2 - 2x + 4y \leq 1$
- B  $x^2 - y^2 - 2x + 4y \leq 1$
- C  $x^2 + y^2 \geq 1$
- D  $xy - z^2 \geq 0$

**Problem 18.** Which of the following matrices is symmetric positive semidefinite?

- A  $\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$
- B  $\begin{bmatrix} 3 & -2 \\ 2 & 4 \end{bmatrix}$

$$\text{C } \begin{bmatrix} -6 & -5 \\ -5 & -3 \end{bmatrix}$$

$$\text{D } \begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix}$$

**Problem 19.** Given a hyperplane  $H : 6x_1 - 8x_2 = -1$  and a point  $p = [-1, 1]$ , find the distance between  $H$  and  $p$ .

A  $\frac{3}{2}$

B  $\frac{2}{3}$

C  $\frac{13}{10}$

D  $\frac{10}{13}$

**Problem 20.** How many possible values can an integer variable  $x$  take where  $x \in [0.5, 3.5]^2 \cap \mathbb{Z}^2$ ?

A 4

B 9

C 10

D 16

**Problem 21.** “If items 1 and 3 are chosen, then item 2 is chosen” can be modeled using binary variables  $x_1, x_2, x_3 \in \{0, 1\}$  as:

A  $x_2 \leq x_1, x_2 \leq x_3$

B  $x_2 \geq x_1 + x_3 - 1$

C  $x_1 \geq \frac{(x_2 + x_3)}{2}$

D  $x_2 \leq x_1 \leq x_3$

**Problem 22.** “If item 2 is chosen, then item 1 or 3 is chosen” can be modeled using binary variables  $x_1, x_2, x_3 \in \{0, 1\}$  as:

A  $x_2 \leq x_1, x_2 \leq x_3$

B  $x_2 \leq x_1 + x_3$

C  $x_2 \leq x_1, 1 - x_2 \leq x_3$

D  $x_2 \leq x_1, 1 - x_1 \leq x_3$

**Problem 23.** “If item 2 is not chosen, then item 1 and item 3 are chosen” can be modeled using binary variables  $x_1, x_2, x_3 \in \{0, 1\}$  as:

A  $x_2 \leq x_1 \leq x_3$

B  $x_1 + x_2 \leq 1, x_2 + x_3 \leq 1$

C  $x_1 + x_2 \geq 1, x_2 + x_3 \geq 1$

D  $x_2 \leq x_1, x_2 \leq x_3$

**Problem 24.** “If item 2 or item 3 is chosen, then item 1 is chosen” can be modeled using binary variables  $x_1, x_2, x_3 \in \{0, 1\}$  as:

A  $x_1 \leq x_2 + x_3$

B  $x_1 \leq x_2 \leq x_3$

C  $x_1 \geq x_2, x_3 \geq x_1$

D  $2x_1 \geq x_2 + x_3$

**Problem 25.** “Either  $x_1 + 3x_2 \leq 3$  or  $4x_1 - x_2 \geq 1$ ” can be formulated as  $y \in \{0, 1\}$ ,  $x_1 + 3x_2 \leq 3 + My$  and

A  $4x_1 + x_2 \geq 1 + My$

B  $x_1 - x_2 \leq 2 - M(1 - y)$

C  $4x_1 - x_2 \geq 1 - M(1 - y)$

D  $4x_1 - x_2 \geq M(1 - y)$

**Problem 26.** “If  $5x_1 + x_2 > 5$ , then  $x_1 - 2x_2 + x_3 \geq 8$ ” can be formulated as  $y \in \{0, 1\}$ ,  $x_1 - 2x_2 + x_3 \geq 8 - M(1 - y)$  and

A  $5x_1 + x_2 \geq 5 + M(1 - y)$

B  $5x_1 + x_2 \leq 5 + My$

C  $5x_1 + x_2 > 5 - M(1 - y)$



**Problem 27.** Consider the assignment problem

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^m x_{ij} = 1, \quad j = 1, \dots, n, \\ & \sum_{j=1}^n x_{ij} \leq 1, \quad i = 1, \dots, m, \\ & x_{ij} \geq 0, \quad i = 1, \dots, m, \quad j = 1, \dots, n \end{aligned}$$

where  $i$  is the index for workers and  $j$  is the index for jobs. If we are going to model this problem as a network flow problem, how many nodes are supply nodes and how many nodes are demand nodes?

- A  $m + 1$  supply nodes and  $n$  demand nodes
- B  $n$  supply nodes and  $m$  demand nodes
- C  $n$  supply nodes and  $m + 1$  demand nodes
- D  $m$  supply nodes and  $n + 1$  demand nodes

**Problem 28.** The number of possible tours in a TSP with 4 cities is at most

- A 24
- B 1024
- C 6

**Problem 29.** We have a set of surgeons and a set of surgery types. Each surgeon can do a subset of the surgery types. We need to pick a subset of surgeons such that, for each surgery type, there is at least one surgeon in the selected set that can do it. This is an example of a

- A Set packing problem
- B Set covering problem
- C Set partitioning problem

**Problem 30.** Which of the following statements is true?

- A A minimum cost network flow problem with integer data with an optimal solution has an integer optimal solution
- B A minimum cost network flow problem always has an optimal solution
- C A minimum cost network flow problem with arbitrary data always has an integer optimal solution